Marmara University

Engineering Faculty, Electrical and Electronics Engineering

EE4077 Fundamentals of Machine Learning, Midterm Exam Solutions, Spring2026

1. (a) $X=\begin{bmatrix}1 & 1^2\\ 1 & 2^2\\ 1 & 3^2\end{bmatrix}=\begin{bmatrix}1 & 1\\ 1 & 4\\ 1 & 9\end{bmatrix}$ (2); $y=\begin{bmatrix}3\\5\\8\end{bmatrix}$ (2); $w=\begin{bmatrix}w_0\\w_1\end{bmatrix}$ (1)

(b) $X^TX=\begin{bmatrix}3 & 14\\ 14 & 98\end{bmatrix}$, $(X^TX)^{-1}=\frac{1}{98}\begin{bmatrix}98 & -14\\ -14 & 3\end{bmatrix}$ (5); $X^Ty=\begin{bmatrix}16\\95\end{bmatrix}$ (5), $w^*=(X^TX)^{-1}X^Ty=\begin{bmatrix}2.428\\0.622\end{bmatrix}$ (5)

(c) **True** because there is only one solution for w^* when X^TX is non-singular (5)

2. (a) $P(Y=1)=3/5, P(Y=0)=2/5.$ (1)

$Y=1 \Rightarrow P(X_1=1|Y=1)=1, P(X_2=1|Y=1)=2/3.$ (2) (2)

$Y=0 \Rightarrow P(X_1=1|Y=0)=0, P(X_2=1|Y=0)=1/2.$ (2) (2)

(b) With Laplace smoothing (k = 1):

$P(Y=1)=(3+1)/(5+2)=4/7, P(Y=0)=(2+1)/(5+2)=3/7.$ (3)

$P(X_1=0|Y=1)=(0+1)/(3+2)=1/5, P(X_2=1|Y=1)=(2+1)/(3+2)=3/5.$ (3)

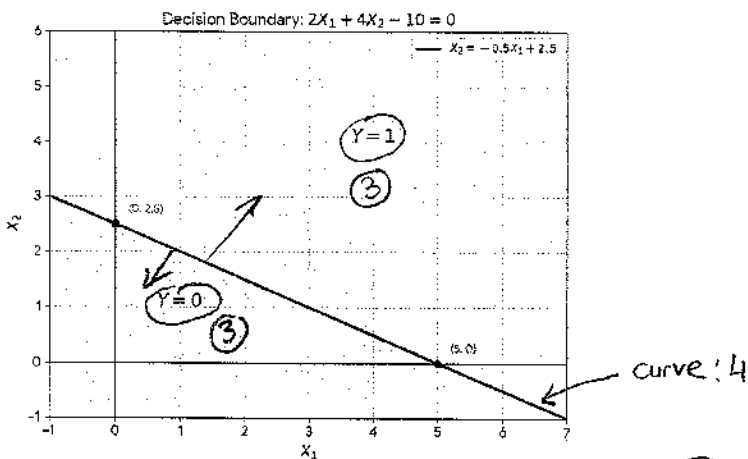
$P(X_1=0|Y=0)=(2+1)/(2+2)=3/4, P(X_2=1|Y=0)=(1+1)/(2+2)=2/4=1/2.$ (3)

Score for $Y=1$: $(4/7)\cdot(1/5)\cdot(3/5)=12/175\approx 0.068.$ (2)

Score for $Y=0$: $(3/7)\cdot(3/4)\cdot(1/2)=9/56\approx 0.16.$ (2)

Result: **Fail (Y=0)** (2)

3. (a)



curve : 4

(b) $z=2(2)+4(2)-10=4+8-10=2.$ (5)

$P(Y=1)=\sigma(2)=\frac{1}{1+e^{-2}}\approx\frac{1}{1+0.135}\approx 0.88$ (5)

4.

(a) **False.** High accuracy on training data often indicates **overfitting**, which can lead to poor performance on unseen test data. (5)

(b) **False.** It uses single samples (or mini-batches), the error might fluctuate and is not guaranteed to decrease at every step. (5)

(c) **False.** It is a classification method. Standard regression for continuous values uses different techniques. (5)

(d) **True.** Using the test set to tune parameters makes it part of the training process, preventing it from being an unbiased measure of generalization. (5)

(e) **True.** Hard-margin SVM requires a hyperplane that perfectly separates all points; if no such plane exists, the optimization fails. (5)